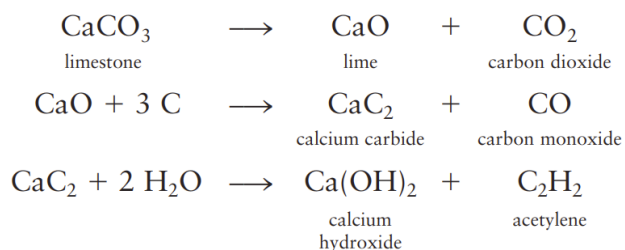


1. Iodine (I) and fluorine (F) form a series of binary compounds with the following compositions:

| Compound | Mass% I | Mass% F |
|----------|---------|---------|
| 1        | 86.979  | 13.021  |
| 2        | 69.007  | 30.993  |
| 3        | 57.191  | 42.809  |
| 4        | 48.829  | 51.171  |

- (a) Compute in each case the mass of fluorine that combines with 1.0000 g iodine.
- (b) By figuring out small whole-number ratios among the four answers in part (a), show that these compounds satisfy the law of multiple proportions.
2. Gaseous methanol ( $\text{CH}_3\text{OH}$ ) reacts with oxygen ( $\text{O}_2$ ) to produce water vapor and carbon dioxide. What volumes of water vapor and carbon dioxide will be produced from 2.0 L methanol if all gases are held at the same temperature and pressure conditions?
3. Determine the mass (in grams) of exactly 100 million atoms of fluorine if the relative atomic mass of fluorine is 18.998403 on a scale on which exactly 12 is the relative atomic mass of  $^{12}\text{C}$
4. An industrial-scale process for making acetylene consists of the following sequence of operations:



What is the percentage yield of the overall process if 2.32 metric tons  $\text{C}_2\text{H}_2$  is produced from 10.0 metric tons limestone (1 metric ton = 1000 kg)?

5. A gold nucleus is located at the origin of coordinates, and a helium nucleus initially at the coordinates  $(0, 2 \text{ \AA})$  moves to the position  $(0, 1 \text{ \AA})$ .
- (a) Calculate the magnitude of the force on the electron at each separation.
  - (b) Calculate the change in the potential energy of the two nuclei.
  - (c) Calculate the change in the kinetic energy of the helium nucleus assuming its value at  $2 \text{ \AA}$  is  $2.0 \text{ MeV}$  if you assume the gold nucleus is so massive that it stays at the origin.